

Context Graph Demo Design Document

Purpose

Demonstrate how Context Graphs enable AI agents to preserve decisions, reasoning, history, and memory across interactions.

The goal is to show the difference between:

- Information retrieval (RAG)
- Context preservation (Context Graphs)

The audience should conclude:

"AI agents do not fail because information is missing. They fail because context is missing."

Demo Scenario

Buying a House

A user is evaluating multiple homes and using an AI assistant to help make a decision.

The assistant must not only compare houses but also remember:

- Preferences
- Decisions
- Alternatives considered
- Reasoning behind decisions
- Historical interactions

This mirrors enterprise decision-making while remaining relatable to all audiences.

Business Objective

Show that the AI agent can answer:

"What happened?"

AND

"Why did it happen?"

AND

"What should we do next?"

using preserved context.

Demo Flow

Step 1 — Initial Evaluation

User

Help me compare these three houses.

Available Houses

House A

Price: \$425,000

Schools: 4/5

Commute: 4/5

Taxes: 2/5

Crime Rate: 4/5

Resale Value: 4/5

HOA: 3/5

House B

Price: \$475,000

Schools: 5/5

Commute: 2/5

Taxes: 4/5

Crime Rate: 4/5

Resale Value: 2/5

HOA: 2/5

House C

Price: \$399,000

Schools: 3/5

Commute: 5/5

Taxes: 4/5

Crime Rate: 4/5

Resale Value: 4/5

HOA: 5/5

Agent Response

After comparing all criteria:

Recommended:

- House A
- House C

Concerns:

- House B has long commute
 - Higher HOA impact
 - Lower resale outlook
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Step 2 — Explicit Decision

User

Let's remove House B from consideration.

Agent

House B removed from active consideration.

Reason:

- Commute exceeds preferred threshold
 - HOA costs are higher than desired
 - Resale potential ranks lowest among evaluated options
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Context Graph Update

Create Decision Node

Decision: Reject House B

Alternatives Considered: House A House C

Reasoning: Commute HOA Resale

Timestamp: Current Date

Decision Status: Completed

Step 3 — Return Later

Simulate one week later.

User

Which houses are we currently considering?

Agent

Active Candidates:

- House A
- House C

Previously Rejected:

House B

Reason:

- Long commute
- Higher HOA costs
- Lower resale ranking

Decision Date: March 10

Step 4 — The Key Question

User

Why did we reject House B?

Expected Agent Response

House B was rejected because:

Primary Factors:

1. Commute exceeded desired threshold
2. HOA costs exceeded target range
3. Resale outlook ranked lowest

Alternatives Considered:

House A House C

Decision Date:

March 10

Decision Rationale:

Long-term affordability and commute convenience were prioritized over school ranking improvements.

Demonstrated Context Types

Information

Relationships

History

Decisions

Reasoning

Step 5 — Introduce New House

New House D

Price: \$435,000

Schools: 4/5

Commute: 5/5

Taxes: 4/5

Crime Rate: 4/5

Resale Value: 5/5

HOA: 4/5

User

Should we consider House D?

Agent

Yes.

House D aligns with the criteria that influenced previous decisions.

Advantages:

- Better commute
- Lower HOA burden
- Strong resale outlook

Recommendation:

Add House D to active consideration.

Step 6 — Reasoning Validation

User

Why are you recommending House D?

Agent

The recommendation is based on previously established priorities.

Key Priorities:

1. Commute under target threshold
2. Lower HOA costs
3. Strong resale potential

These priorities were derived from the decision to reject House B.

House D satisfies all three conditions.

Context Graph Model

Entity Types

House

User

Decision

Reason

Preference

Evaluation

Conversation

Recommendation

Relationships

User -> EVALUATED -> House

User -> REJECTED -> House

Decision -> CONSIDERED -> House

Decision -> BASED_ON -> Reason

Recommendation -> GENERATED_FROM -> Decision

House -> HAS_ATTRIBUTE -> Commute

House -> HAS_ATTRIBUTE -> HOA

House -> HAS_ATTRIBUTE -> Resale

Context Stored

Information

House attributes

Relationships

House comparisons

Decision dependencies

History

Timeline of evaluations

Decision timestamps

Decisions

Accepted

Rejected

Deferred

Reasoning

Trade-offs

Decision rationale

Priority rankings

Demo UI Recommendation

Three-panel layout.

Left Panel

Chat Interface

User questions

Agent responses

Center Panel

Current Evaluation

House comparison cards

Current recommendations

Right Panel

Context Timeline

Decision History

Reasoning

Preferences

Context Graph Events

What Traditional RAG Would Do

Can retrieve:

- House descriptions
- School information
- HOA documents
- Tax records

Cannot reliably answer:

Why was House B rejected?

What priorities influenced that decision?

What decision was made last week?

What reasoning led to the recommendation?

What Context Graph Enables

Persistent memory

Decision continuity

Reasoning traceability

Explainable recommendations

Context-aware decision making

Demo Success Criteria

The audience should leave with one key insight:

The value is not that the AI found better information.

The value is that the AI remembered the decision, preserved the reasoning, and reused that context later.

That is the role of a Context Graph.